

Advanced (Habanero)

Q1. Convert $x^2 + 6x + 8$ to vertex form.

- (A) $(x + 3)^2 - 1$
- (B) $(x + 4)^2 - 8$
- (C) $(x + 3)^2 + 1$
- (D) $(x + 2)^2 + 4$
- (E) $(x + 1)^2 - 7$

Q2. Complete the square for $x^2 + 4x$.

- (A) $(x + 2)^2 - 4$
- (B) $(x + 1)^2 - 1$
- (C) $(x + 3)^2 - 9$
- (D) $(x + 4)^2 - 16$
- (E) $(x + 2)^2$

Q3. What is the vertex of the parabola $x^2 + 4x + 4$?

- (A) $(-2, 0)$
- (B) $(-1, -1)$
- (C) $(-4, 4)$
- (D) $(-3, 1)$
- (E) $(-2, 4)$

Q4. Convert $x^2 - 8x + 12$ to vertex form.

- (A) $(x - 4)^2 - 4$
- (B) $(x - 3)^2 + 3$
- (C) $(x - 2)^2 + 8$
- (D) $(x + 4)^2 - 4$
- (E) $(x - 1)^2 - 11$

Q5. What is the equation of the parabola with vertex $(-2, 3)$ and x^2 coefficient of 1?

- (A) $(x + 2)^2 + 3$
- (B) $(x - 2)^2 - 3$
- (C) $(x + 1)^2 + 2$
- (D) $(x + 2)^2 - 3$
- (E) $(x - 2)^2 + 3$

Q6. Complete the square for $x^2 - 6x$.

- (A) $(x - 3)^2 - 9$
- (B) $(x + 3)^2 - 9$
- (C) $(x + 2)^2 - 4$
- (D) $(x - 2)^2 - 4$
- (E) $(x - 1)^2 - 1$

Q7. What is the vertex of the parabola $x^2 + 2x + 1$?

- (A) $(-1, 0)$
- (B) $(-2, 0)$
- (C) $(-1, -1)$
- (D) $(-2, 4)$
- (E) $(-1, 1)$

Q8. Convert $x^2 - 2x - 6$ to vertex form.

- (A) $(x - 1)^2 - 7$
- (B) $(x + 1)^2 + 7$
- (C) $(x + 2)^2 - 2$
- (D) $(x - 2)^2 - 2$
- (E) $(x + 1)^2 - 5$

Open Ended Questions (FRQ)**Q9.** Find the vertex and x -intercepts of the parabola $x^2 + 5x + 6$. Show all work.**Q10.** Write the equation of the parabola with x^2 coefficient of 1, vertex $(-3, 2)$, and passing through point $(1, 4)$. Show all work.**Chef's Tips**

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- Tip 1: Make sure to check for extraneous solutions when completing the square.
- Tip 2: Have students show all work and explain their reasoning.
- Tip 3: Emphasize understanding of vertex form and its applications.